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1 Realistic Power System State Estimator
2
3 Group Members:
4 1. RANA MUHAMMAD ADNAN KHALIL
5 2. EREN DENİZ EMİNAĞAOĞLU
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7 Course: EE574
8 Instructor: MURAT GÖL
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10 Objective:
11 Understand the working of Power System State Estimator using WLS and observe the
12 behavior of estimator with different types of data set.
13
14 Scope:
15 Parsing of IEEE format network data file.
16 Parsing of measurements file.
17 Determination of:
18 i) Bus admittance matrix (ybus).
19 ii) Measurement Vector
20 ii) Jacobian matrix
21 iii) Gain matrix
22 iv) state estimator using WLS
23 v) Observability analysis
24 vi) Bad data detection
25
26 Pro and Cons of Project
27 Pro:
28 This code can handle 8 type of measurements
29 1. Voltage magnitude
30 2. Voltage angle
31 3. Active Power Injection
32 4. Reactive Power Injection
33 5. Active Power Flow
34 6. Reactive Power Flow
35 7. Current Magnitude
36 8. Current Angle
37 This code can handle missing measurements
38 This code can handle noisy measurements
39 Self explanatory output
40 This code can filter measurements which are not in the network.
41 The code is not dependent on fixed files, it asks user to enter file name of cdf and
42 measurement file respectively.
43
44 Cons:
45 It assumes the cdf file should be correct.
46
47 AI Reference:
48 The code skeleton is created with the help of AI tool Claude.
49
50 How to The Project:
51 First install the PyCharm Community Edition.
52 This project is build in open source PyCharm Community Edition, the detail of version
53 is given below
54 PyCharm 2025.2.6.1 (Community Edition)
55 Build #PC-252.28539.58, built on April 21, 2026
56 Source revision: af37552c2b565
57 Runtime version: 21.0.9+1-b1038.78 amd64 (JCEF 122.1.9)
58 VM: OpenJDK 64-Bit Server VM by JetBrains s.r.o.
59 Toolkit: sun.awt.windows.WToolkit
60 Windows 10.0
61 GC: G1 Young Generation, G1 Concurrent GC, G1 Old Generation
62 Memory: 1500M
63 Cores: 4
64 Registry:
65     ide.experimental.ui=true
66     llm.show.ai.promotion.window.on.start=false
67
68 Project Dependencies:
69 After installing PyCharm Community Edition, open the project and install the
70 following modules.
71 Go go settings then interpreter click '+' sign and add the following modules and
72 install.
73 numpy
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69  scipy
70  tabulate
71  matplotlib
72  networkx
73  pip
74  pyparsing
75
76  Run the project by pressing shift + F10 key.
77  If there is any other dependent module then the error message will appear with that
78  name and install that module as well.
79  On successful run, the output window ask the cdf file name, please enter the correct
80  name and the file should be in project folder.
81  After that the output window will ask the name of measurement file, please enter the
82  correct name and the file should be in project folder.
83
84  If the network is observable nad gain matrix is non-singular then ac-wls algorithm
85  will run smoothly.
86  Otherwise ac-wls will stop but the program will try to run bad data analysis.
87  Read the output window carefully to see the step by step output.
88  The output window is self explanatory.
89
90  Notes:
91  1. If the first absolute value of measurements in block 2 is greater than 0.1, it
92  will be considered as power injection measurement block.
93  2. The parsing of measurement file is very important and tricky, all calculations
94  depends on this parsing.
95  3. The program assumes the measurements are synchronized and there is no time skew.
96  4. The program is tested with 6 different types of test cases.
97
98  Graph
99  Figure 1 is residuals graph.
100 Figure 2 shows largest normalized residuals.
101
102 Future
103 Android mobile app is being developed for this state estimator, so that users can get
104 states, observability analysis and bad data detection in their pocket.
105 Beta version is available on google play store.
106 Further features will be added in near future.
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